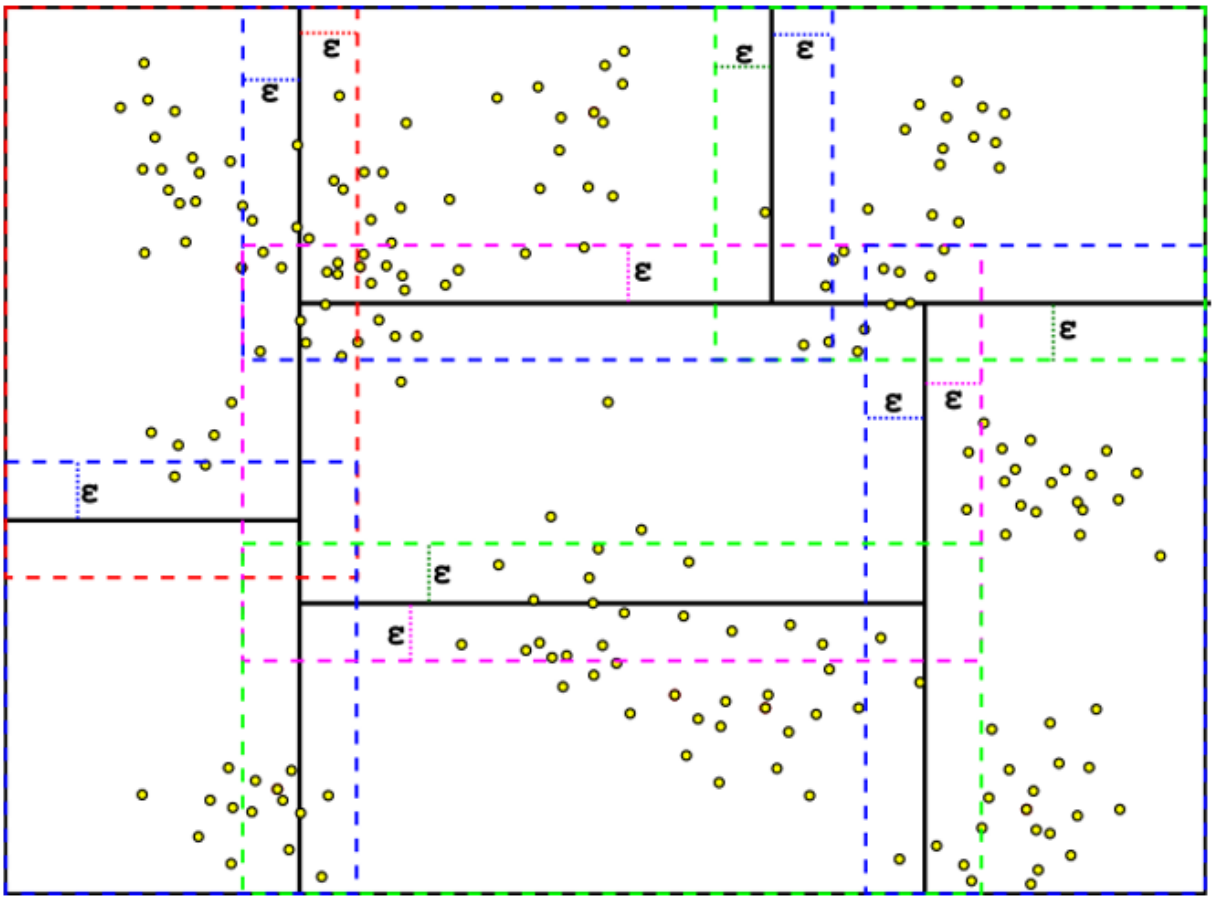


Figure 13.14 An example of an irregularly shaped grid for sharding a dataset in MR-DBSCAN. Notice that shards (dashed-lines rectangles) are built by expanding cells in each direction for a distance equal to ε, the dense region's radius.

To speed things up, we can use an R-tree: as mentioned in chapter 10, R-trees can hold non-zero-measure objects, and in particular shapes like rectangles (see figure 10.5) – we can therefore create an R-tree whose items are the grid cells, and for each point find the closest cell. Since R-trees have linear-time worst-case running time, however, we don't improve our asymptotic result (but in most cases R-tree will result, in practice, faster than naïve search).